

Pennada Venkata Chaitanya

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Professional Summary

AI/ML Software Engineer with **2.5+ years of hands-on experience** building production Generative AI systems, AI agents, RAG applications, distributed backend services, and scalable data pipelines. Experienced in Python, FastAPI, LLMs, LangGraph, MCP, vector search, PostgreSQL, MongoDB, and production AI automation.

Professional Experience

Software Engineer | Innocito

April 2025 – Present

- Engineered an AI-powered fashion discovery platform, processing and mapping **5 million+ fashion product records** through high-throughput AI pipelines for large-scale preprocessing, classification, and catalog enrichment.
- Designed and implemented **distributed data pipelines** using Python, Prefect, FastAPI, and MongoDB, leveraging asynchronous processing to automate production workflows and reduce processing time from **14 days to 1 day**.
- Created a real-time **AI Voice Agent** integrating LLMs, speech-to-text, and text-to-speech, reducing end-to-end response latency by **3 seconds per request** and enabling more natural conversations.
- Implemented local and remote **Model Context Protocol** servers for structured tool routing and integration between AI agents and backend services.

AI Software Developer | AI Gen media

March 2024 – February 2025

- Developed AI-driven text-to-video workflows, achieving **80% visual consistency** on predefined evaluation benchmarks.
- Built an automated AI GIF Studio, achieving **95% test-case accuracy**, **5x faster runs**, and **10,000+ interactions**.
- Automated social media workflows using AI agents, reducing manual workflow time by **30%**.

Projects

Synapse

[Website](#) | [GitHub](#)

- Architected an **offline-first AI knowledge system** spanning **4 core capabilities**: block-based knowledge management, information retrieval, task orchestration, and agent-driven actions.
- Designed **dual-database synchronization** between SQLite and PostgreSQL using block-level incremental updates and Hybrid Logical Clock (HLC)-based conflict resolution.
- Engineered **hybrid local-cloud Retrieval-Augmented Generation (RAG)** pipelines with embedding generation, incremental vector indexing, semantic retrieval, context assembly, and citation-grounded LLM response generation.
- Integrated **LangGraph-based agent orchestration** with MCP tool execution and local/cloud LLM workflows, enabling retrieved knowledge to drive structured actions while supporting local-first AI execution.

Tech: Python, FastAPI, PostgreSQL, SQLite, LangGraph, Ollama, MCP, Vector Search.

Skills

Programming: Python, SQL, C++

Generative AI & Agents: Large Language Models (LLMs), Multimodal LLMs, AI Agents, Retrieval-Augmented Generation (RAG), Vector Search, Model Context Protocol (MCP), LangGraph, LangChain, LlamaIndex, AI Observability, Ollama

Machine Learning & AI: PyTorch, NLP, Computer Vision, Model Evaluation, Data Preprocessing, Feature Engineering

Backend & Data: REST APIs, FastAPI, PostgreSQL, MongoDB (NoSQL), SQLite, Prefect, Asynchronous Processing

Engineering Tools: Pytest, Playwright, Git, GitHub, Docker, Vast AI, Jira, Agile/Scrum, Automation, Streamlit

Achievements

- Won the institutional round of the **Solving for India Hackathon**, hosted by Google, AMD, and GeeksforGeeks on [Devfolio](#), and the international **Fresh Hacks** hackathon hosted on [Devpost](#).
- Received the **Rising Star Award** at Innocito for strong performance, rapid growth, and technical contributions.
- Solved **300+ DSA problems** across [LeetCode](#) and [GFG](#). Achieved a **5-star Problem Solving** rating on [HackerRank](#).

Education

B.Tech in Computer Science | Aditya College of Engineering and Technology

2020 – 2024

Focus: Artificial Intelligence & Machine Learning